

Algebra II with Trigonometry

Chapter 6.3

Olympiad Problems (GPT-5.4)

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[4531393_alg-2trig-h-chp-6-3-note.pdf](#)

Problems

1. Find the exact maximum value of

$$\sin x + \cos x + \sin 2x$$

for real x .

2. A sinusoid of the form

$$y = a \cos(kx - b) + c$$

has one complete period on $[1, 5]$, maximum value 7, minimum value -1 , and passes through $(1, 3)$ and $(3, 7)$. Determine one such equation.

3. Compute exactly:

$$\sin \frac{\pi}{12} + \sin \frac{5\pi}{12} + \sin \frac{9\pi}{12} + \sin \frac{13\pi}{12} + \sin \frac{17\pi}{12}.$$

4. Let

$$f(x) = x - 2 \sin x \quad (0 \leq x \leq 2\pi).$$

Without calculus, determine how many solutions the equation $f(x) = \pi$ has, and locate each solution exactly if possible.

5. Find the least positive period of

$$g(x) = \sin\left(\frac{\pi x}{2}\right) + \cos\left(\frac{\pi x}{3}\right).$$

6. For real x , define

$$h(x) = \cos x + \cos\left(x + \frac{2\pi}{3}\right) + \cos\left(x + \frac{4\pi}{3}\right).$$

Evaluate $h(x)$, and then use the same idea to evaluate

$$\sin x + \sin\left(x + \frac{2\pi}{3}\right) + \sin\left(x + \frac{4\pi}{3}\right).$$